



Spatial Truth: Location Service for Physical AI

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Connectcast Telco Announces "Spatial Truth": A Software-Defined Location Service to End Urban and Rural Navigation Blind Spots.

Amazon and Toyota achieve centimeter-accurate autonomy via software-defined network intelligence - zero hardware required.

Today, Connectcast announced **Spatial Truth**, a software-defined **Network-as-a-Sensor** location service that gives enterprise vehicle fleets **centimeter-level accuracy** with zero additional hardware. Spatial Truth is built for large-scale operators, such as Amazon and Toyota, that require reliable, lane-level accuracy across warehouses, ports, yards, logistics hubs, and public roadways.

The service eliminates location blind spots. In warehouses and ports, vehicles experience constant signal bounce-offs from steel structures, containers, cranes, and stacked inventory, creating large multipath errors that distort positioning. In dense urban environments, skyscrapers, glass façades, and metal infrastructure cause GPS reflections and interference that routinely generate 10-20-meter inaccuracies. And across rural open-road routes, vehicles and autonomous systems encounter complete signal dropouts, leaving navigation systems unable to determine even coarse location.

Spatial Truth resolves these blind spots by shifting accuracy from the vehicle to the network. Using only the standard radios such as 5G, Wi-Fi RTT, and terrestrial signals already built into today's connected vehicles. Spatial Truth fuses these inputs inside the 5G core to create a software-defined "source of truth" for location. The result is reliable, centimeter-level accuracy delivered at fleet scale, with zero hardware.

Today, approximately 72 million connected vehicles operate under conditions where GPS becomes unreliable or unusable. Warehouses, ports, and dense city centers produce multipath distortion; glass and steel surfaces generate false reflections; and rural areas experience full GPS signal loss. These errors cause lane-level navigation errors, missed ETAs, routing mistakes, slowed automation, and safety exposures, for fleets at scale. Of these **72 million** vehicles, we focus on a serviceable market of **45 million** already on 5G networks, where blind-spots drive **\$3.24B** annual impact.

Across enterprise fleets, these location errors collectively generate an estimated \$3.24B annual P&L drag. Historical solutions, such as adding LiDAR, RTK modules, or other high-cost sensors, cost **\$5,000+ per vehicle**, making autonomy expensive to deploy widely.

EXECUTIVE PERSPECTIVE "For years, connectivity and location were treated as silos. Spatial Truth proves the network *is* the sensor," said the VP of IoT Engineering at Cisco. "By enabling this through the Cisco IoT Control Center, the #1 platform for ~300 million connected devices, *we stop selling megabits and start selling guaranteed centimeter-level location.*"

90-DAY ACTION PLAN:

Proposed plan for the enterprise to validate the proof of concept:

1. Days 1-30: Audit current LMF/Core architecture and align with Toyota and Amazon on dead-zone pain points.
2. Days 31-60: Execute a PoC using existing gNB hardware to demonstrate centimeter accuracy in a simulated signal bounce-off environments.
3. Days 61-90: Launch Location Service beta tier, converting connectivity into high-margin SaaS location service.

Sources

Statista – [Connected Cars Worldwide : 400M+ vehicles projected by 2025](#)

5G Automotive Association (5GAA) – [Over 300M vehicles with advanced connectivity today](#)

Juniper Research/ Business Standard – [Commercial vehicle ~20% of connected vehicle base by 2027.](#)

Internal FAQs

Q 1) What is the customer demand and market opportunity for Location Service?

(Appendix A: Market Projections and SAM)

A: In consumer space the success of Life360 and its 80 million subscribers proves: people pay for location accuracy.

Consumers already expect continuous location accuracy, even when the network fails. That's why Apple built Emergency SOS, and why Skylo provides NTN continuity in rural dead zones. These technologies didn't *optimize* uncertainty—they removed it.

Spatial Truth brings that same leap to the enterprise: It eliminates the root problem, not the symptoms. Without it, commercial vehicle fleets keep paying the "location uncertainty tax"—every hour, every route, every day.

We define the market not by data usage, but by the active connected vehicle base. Our Serviceable Addressable Market (SAM) consists of **45M** commercial vehicles globally that currently pay a "location uncertainty tax."

- The opportunity at a **\$6/month blended ARPU**, is **\$3.24B**
- The annual projection: With a conservative 5% penetration of this connected vehicle base (2.25M vehicles X \$6/month X 12 months), Spatial Truth projects \$162M in Annual Recurring Revenue (ARR).
- The Demand: Much like Life360 proved consumers will pay for location certainty, enterprises like Amazon and Toyota require a centimeter-accuracy SLA to scale operation efficiency. Spatial Truth provides this through three subscription tiers:
 - \$3/month for <1m (Bronze)
 - \$7/month for <30cm (Silver)
 - \$15/month for <10cm (Gold)
- Exclusions for Conservatism: The 45M figure excludes consumer passenger vehicles (despite Life360's 80M+ subscriber proof-of-concept), emerging 5G markets, and non-automotive sectors like maritime or agriculture.

Q 2) Why is the monthly subscription of, "\$3 / \$7 / \$15," per month the right price point?

(Appendix B: Pricing and Financial Projection Deep Dive)

A: The \$6.00 blended monthly ARPU represents a shift from selling low-margin "megabits" to high-margin Location Service. For a truck generating \$200k/year in revenue, a \$15/month (\$180/year) "Gold" tier represents just 0.09% of operating costs. Payback Calculation: If our location accuracy prevents just one 25-minute warehouse delay per month, the subscription pays for itself (\$204 in annual recovered time vs. \$180 annual cost if priced at \$15/month):

- 25 min delay = ~0.42 hours.

- \$40/hour burdened driver cost = 0.42 hours X \$40/hour = \$17 per incident/month.
- 12 months X \$17 incidents/month = \$204 per vehicle in annual waste.

Conclusion: A single avoided urban city missed turn or warehouse dwell incident per month pays the entire annual investment.

Q3) How does this impact the P&L of a 1,000-vehicle enterprise fleet? (*Appendix C: Financial Projection for 1000-fleet vehicle*)

A: Location error is a direct P&L drag. For a 1,000-vehicle fleet generating \$180M–\$200M in revenue, Spatial Truth recovers \$11M–\$29M annually in location uncertainty tax.

- \$1.2M–\$3.6M/year in routing efficiency by eliminating 1-3% GPS navigation loss.
- \$800K–\$1.5M/year in yard efficiency by reducing warehouse/yard inefficiency by 8-15 min./stop).
- 10-20% Reduction in insurance premiums by providing a certified “source of truth” for liability.

Table 1.0: Location uncertainty tax recovers \$11M - \$29M annually for 1,000-Vehicle Fleet

Use Case	Stakeholder	The Problem	The Impact	The Recovery
#1 Routing	Fleet Manager	10–20m drift causes missed turns and circling.	1-3% inefficiency; \$1.8M–\$5.4M gross loss.	\$1.2M–\$3.6M in direct waste recovered.
#2 Yard Dwell	Logistics Lead	Poor fidelity adds 8-15 mins of manual dwell per stop.	500 fleet-hours lost daily; \$6M+ annual waste.	\$800K–\$1.5M saved via automated check-ins.
#3 Utilization	COO	Unreliable telemetry prevents confident job reassignment.	5-12% utilization drag; ~\$9,000 lost per truck.	\$9M–\$21.6M utilization loss.
#4 Insurance	Safety Officer	GPS "jitter" creates ambiguous liability evidence.	10-20% premium hikes on \$6.5K–\$10K base costs.	\$650K–\$2M reduction in annual premiums.
#5 Autonomy	CTO	GPS drift forces disengagements and \$5K+ hardware fixes.	Frozen CAPEX; autonomy programs stuck in test loops.	Accelerated ROI; scales L2.5/L3 to urban routes.

Q 4) What real customer problem does Spatial Truth solves?

A: Enterprises currently pay a "location uncertainty tax." Location errors of 10-20 meters in urban cities, lane-level inaccuracies in warehouses, and rural signal dropouts lead to missed ETAs, routing errors, and safety exposures.

Spatial Truth solves this by addressing three critical friction points:

- Eliminates the "Integrator's Burden": Traditional accuracy requires thousands in upfront CAPEX per vehicle for hardware (LiDAR/RTK), followed by months of installation. Spatial Truth is a Zero-CAPEX software update that scales across global fleets instantly via the Cisco IoT Control Center.
- Predictable Automation: Spatial Truth moves location from being a "best-effort" utility to a high-accuracy sensor, enabling L2.5/L3 autonomy to function reliably in dense urban or indoor environments where GPS currently fails.
- The "Google Maps" Moment for Enterprise: Spatial Truth provides the same leap in certainty that consumers felt when trip-planning became effortless. The same "peace of mind" that Life360 provides families.

Q7) Who should build this?

A: The solution requires orchestration across four key layers:

- Infrastructure Providers (The Backbone): Companies like Cisco, Ericsson, and Huawei. They provide the 5G Standalone (SA) cores and Location Management Functions (LMF), where the AI layer is hosted. *The Advantage*: These players own the raw signal data, allowing them to use reflection signals to sharpen accuracy—a capability standalone GPS chips lack.
- Platform Orchestrators (The Scale): Platforms like the Cisco IoT Control Center. They provide access to the ~300 million connected devices.
- Global Constellations (The Continuity): Partners like Skylo provide the NTN (Non-Terrestrial Network) bridge for rural continuity.
- The Enterprise Fleets (The Demand): Large-scale operators like Amazon and Toyota that provide the high-volume, recurring revenue data.

Q 6) Why should the build? What leverage do they have?

A: Ownership of the raw network signal. Unlike standalone GPS, Spatial Truth uses reflections as "signal fingerprints" to sharpen accuracy where others see noise. Spatial Truth doesn't optimize uncertainty; it removes it.

Q 7) How does it work? (*Appendix D: Technical Overview*)

A: Spatial Truth delivers centimeter-level accuracy by combining AI-driven signal processing with seamless terrestrial satellite continuity inside the 5G network core.

- Super-Resolution AI (Physics to ROI): Spatial Truth hosts AI-powered Zainar as a containerized software module inside the 5G LMF (Location Management Function). Instead of treating reflections as noise, the AI interprets them as “fingerprints,” producing centimeter-level location accuracy.
- The Skylo Bridge: For rural continuity, Spatial Truth utilizes NB-NTN standards to trigger a seamless handover from 5G to GEO satellites via the S8 interface, ensuring location visibility in rural areas.
- Deep Integrated Positioning: Spatial Truth network blends 5G and Wi-Fi RTT (802.11mc passive ranging) at the control plane, ensuring a drone navigates safely indoors, around building, or on balconies, without requiring private Wi-Fi access.

APPENDICES FAQ (DEEP DIVES)

Appendix A: Market Sizing & Serviceable Addressable Market (SAM)

Executive Summary By 2027, the global connected vehicle market will reach 360M units. Spatial Truth targets the **72M** commercial vehicles within this base that require centimeter location accuracy. Our Serviceable Addressable Market (SAM) consists of **45M** vehicles operating in mature 5G Standalone (SA) regions, including North America, Western Europe, and East Asia. Where network architecture supports centimeter-level location accuracy.

Q A1: What is the total addressable market (TAM) for location services?

A: We define the market by the active connected vehicle base rather than traditional connectivity ARPU. Connectivity is a "race to the bottom." Companies like Amazon or Toyota want to pay as little as possible for data. If you only sell data, you are just a pipe.

TAM is based on high-value vehicle. The charge is for the fact that the vehicle is now centimeter-accurate regardless of how much data it consumes.

- Global Connected Base: Projected to exceed 400M vehicles by 2025.
- Current Connectivity: The 5G Automotive Association (5GAA) reports 300M+ vehicles with advanced connectivity currently in operation.
- 2027 Consensus: Market analysts project a stabilized base of ~360M-370M connected vehicles globally.

Q A2: How many of these vehicles represent high-value commercial vehicle fleets?

A: Commercial vehicles are the primary drivers of centimeter-level location dependent revenue.

- Commercial Segment: Estimated at 20% of the total connected vehicle base by 2027. (20% of 360M total connected base is ~72M)
- Target Segment: This equates to ~72M connected commercial vehicles globally.
- Strategic Focus: These assets represent the core base addressable by Spatial Truth's accuracy SLAs.

Q A3: What criteria define the Serviceable Addressable Market (SAM)?

A: Our SAM is limited to regions where the network can act as a sensor. This requires:

- 5G Standalone (SA) core network architecture.
- Location Management Function (LMF) software deployment.
- Dense gNB coverage to ensure signal trilateration.
- Regulatory frameworks permitting network-based positioning.

Q A4: Which geographic regions meet these requirements today?

A: Mature 5G SA deployments currently exist in the following high-density logistics hubs:

- Americas/Europe: United States, Canada, and Western Europe.
- Asia-Pacific: Japan, South Korea, and Tier-1 urban markets in China.
- Middle East: GCC countries including the UAE and Saudi Arabia.

Q A5: What is the final SAM calculation?

A: We conservatively estimate that 60% of global connected commercial vehicles operate within the 5G-SA ready regions listed above.

- SAM Calculation: **60% of 72M vehicles = 45M** connected commercial vehicles.
- Exclusions for Conservatism: This figure excludes consumer passenger vehicles (despite Life360's 80M+ subscriber proof-of-concept), emerging 5G markets, and non-automotive sectors like maritime or agriculture.

Q A6: Why is this estimate considered conservative?

A: This sizing ensures a realistic "Day 1" revenue path. It accounts only for regions with active 5G SA readiness and assumes that regulatory approvals are already localized. It ignores massive "early adoption" segments that will likely convert as 5G cores expand globally.

Appendix B: Pricing and Financial Projections

Executive Summary Spatial Truth utilizes a tiered SaaS model to convert centimeter accuracy into recurring revenue. By positioning the service as a low-cost enhancement that removes location uncertainty rather than a primary telematics platform, we align with existing procurement behaviors while maintaining high-margin, zero-CAPEX scalability. SAM projection for **45M vehicles** at \$6/month (\$72/year) blended ARPU is **\$3.24B annual revenue**. At 5% penetration of SAM, the projection for **2.25M vehicles** is **\$162M ARR**.

Q B1: What market benchmarks support the proposed pricing tiers?

A: Our pricing is designed to fit within established enterprise telematics budgets.

Market Context: Commercial telematics subscriptions typically range from \$10-\$40+ per vehicle/month. The "Add-on" Strategy: Spatial Truth is a centimeter accuracy enhancement, not a replacement. Our tiers represent a manageable "fractional" spend:

- Subscription tiers are \$3/month (Bronze): \$7/month (Silver):15/month (Gold).
- Represents ~7-30% of typical telematics spend.
- Positioned as a premium add-on tier.
- High-accuracy mission-critical tier, still within established fleet budget tolerances.

Q B2: What is the economic justification (ROI) for a fleet to pay for these tiers?

A: For a standard commercial truck, the cost of centimeter accuracy is negligible compared to the cost of location error.

- **Operational Baseline:** A truck costs ~\$200,000/year to operate based on ~\$2.00 per mile for 100k miles.
- **The Investment:** The \$180/year, Gold, subscription represents just 0.09% of total operating costs
- **The Payback:** If centimeter accuracy prevents just one 25-minute delay per month (fully burdened driver cost of \$40/hr), the per truck, per incident saves \$204/year.
 - *Driver hours:* 25 minutes = ~0.42 hours.
 - *Incident Cost:* 0.42 hours × \$40/hour = ~\$17 per incident.
 - *Annual Savings:* \$17 × 12 months = **\$204/incident/truck/year**
- **Conclusion:** One avoided delay per truck per month pays for the entire annual subscription.

Q B3: How are the service tiers structured by accuracy and segment?

A: Align value escalation with centimeter accuracy:

Tier	Accuracy	Annual Cost	Target Segment
Bronze	<1m	\$36	Basic fleet/yard operations
Silver	<30cm	\$84	Urban logistics/Last-mile
Gold	<10cm + NTN	\$180	Autonomous-ready/Mission-critical

Logic: Each step in the gradient represents a 3x to 10x improvement in centimeter accuracy

Q B4: Why is this model more capital efficient than traditional hardware solutions?

A: We eliminate the "Hardware Clutter" traditionally required to solve location uncertainty.

- **Legacy Burden:** Traditional high-accuracy solutions require thousands in upfront CAPEX per vehicle, plus installation and maintenance.
- **Spatial Truth Advantage:** By moving to a network-based zero-CAPEX model, a 5-year "Gold" subscription costs only \$900 total.
- **Business Impact:** This shifts the cost from a heavy CAPEX burden to a scalable, high-margin OPEX subscription that can be deployed instantly via software.

Q B5: What revenue does it support at scale?

A: We project significant ARR even with conservative market penetration.

- **Serviceable Addressable Market (SAM):** ~45M vehicles.
- **Blended ARPU:** \$6/month (\$72/year).
- **Total Market Potential:** \$3.24B annual revenue.
- **5% Penetration Target of 45M = 2.25M vehicles: \$162M ARR**

APPENDIX C: Financial Projection for 1000-fleet vehicle enterprise

Executive Summary For an enterprise fleet manager operating 1,000 vehicles and generating \$180M-\$200M in annual revenue, “location uncertainty tax” is **\$11M - \$29M** "P&L drag" on their balance sheet.

UC #1: Routing Inefficiency (\$1.2M - \$3.6M Annual Loss): Location uncertainty causes missed turns, circling blocks, and driver overcompensation when map icons "jump."

- The Friction: 10-20-meter drift in dense urban zones triggers 1–3% in total route inefficiency.
- The Impact: At a fleet revenue baseline of \$180M, this represents a annual loss of \$1.8M - \$5.4M.
- The Recovery: Attributing a ~70% slice to positioning errors, centimeter accuracy recovers \$1.2M-\$3.6M annually in direct routing waste.

UC #2: Yard & Warehouse Dwell (\$800K - \$1.5M Annual Loss): Poor location prevents automated arrivals and dock assignments, inflating "yard dwell" by 8-15 minutes per stop.

- The Friction: 3 stops/day × 10 minutes lost = 500 fleet-hours lost per day.
- The Cost: At \$50-\$75 per vehicle-hour,
- The Daily waste: 500 fleet-hours X \$50 (low end) = \$25,000 per day on wasted time.
- Annual Waste: \$25,000/day X 250 workdays = ~ \$6M/year. (250 workdays per year excluding weekends/holidays).
- The Recovery: By enabling accuracy automated check-ins, Spatial Truth eliminates the 10–15% of dwell time caused by location uncertainty, saving \$800K-\$1.5M per year.

UC #3: Asset Utilization Drag (\$9M - \$21.6M Annual Loss): When telemetry is unreliable, dispatchers cannot reassign jobs with confidence, leaving assets productive for fewer hours per day.

- The Friction: Multi-city fleets report a 5-12% utilization drag due to low location accuracy.
- The Impact: With each truck producing \$180,000/year, a 5% loss equals \$9,000 per truck.
- The Scale: Across 1,000 vehicles, this represents \$9M-\$21.6M in unrealized revenue.

UC #4: Insurance & Liability Exposure (\$650K - \$2M Avoidable Expense): Insurers penalize fleets with "lost telemetry" segments or ambiguous liability evidence caused by GPS jitter during collisions.

- The Friction: Average insurance costs range from \$6,500-\$10,000 per vehicle.
- The Impact: Location uncertainty during safety events contributes to a 10-20% premium increase.

- **The Recovery:** Providing a sub-meter "source of truth" for collision logs can reduce annual fleet premiums by \$650,000-\$2,000,000.

UC #5: Autonomy Stalling (Frozen CAPEX & Delayed ROI): Level 2.5 and Level 3 systems require sub-meter localization for lane-level guidance and safety compliance.

- **The Friction:** When GPS drifts 10-20-meters, autonomous systems disengage, forcing a fallback to expensive "Hardware Clutter" (LiDAR/RTK).
- **The Impact:** This delays the ROI of multi-million-dollar autonomy programs and keeps capital idle.
- **The Solution:** Network-level centimeter accuracy allows these programs to scale beyond test loops into revenue-generating urban routes.

*Table 2.0: Location Error Total Annual Loss for 1,000-Vehicle Fleets **\$11M - \$29M***

Use Case	Low-End Loss	High-End Loss
UC #1: Routing Inefficiency	\$1.2 Million	\$3.6 Million
UC #2: Yard & Warehouse Dwell	\$0.8 Million	\$1.5 Million
UC #3: Asset Utilization Drag	\$9.0 Million	\$21.6 Million
UC #4: Insurance & Liability	\$0.65 Million	\$2.0 Million
TOTAL ANNUAL LOSS	\$11.65 Million	\$28.7 Million

APPENDIX D: Technical Overview

Executive Summary For Centimeter-level accuracy, Spatial Truth uses AI-driven signal reconstruction and tightly integrated 5G–NTN positioning. This appendix outlines how these components work together inside the network core.

Q D1: Architecturally, what is the core shift?

A: Transition from Device-Centric to Network-Centric. Current high-accuracy positioning relies on "Hardware Clutter" (LiDAR/RTK) installed on every asset, creating massive CAPEX and technical debt. Spatial Truth utilizes the Network-as-a-Sensor. The platform processes raw signal data at the edge (the 5G Core) and delivers centimeter accuracy to the device via a unified abstraction layer. This effectively upgrades the positioning capabilities of the network without requiring physical modifications to the vehicle.

Q D2: How is "location uncertainty" resolved in the Urban cities?

A: GPS fails in dense cities because signals bounce off glass and steel. Instead of attempting to filter these reflections as noise, the Spatial Truth AI treats them as a unique environmental signature. Super-Resolution Neural Networks within the LMF (Location Management Function) calculate the direct path among the interference.

The Result: Physics-based interference is transformed into a positioning strength, achieving sub-meter accuracy.

Q D3: How is a "Single API" possible across network vendors (Ericsson, Huawei, Nokia)?

A: Spatial Truth functions as a Northbound Middleware.

- **The Interface:** While the "Southbound" integration requires specific adapters for different Network Function (NF) vendors (Ericsson, Huawei, etc.), Spatial Truth abstracts this complexity.
- **The Benefit:** Enterprise customers (e.g., Amazon, Toyota) interact with a single, standardized API. The platform handles underlying handshakes between the Cisco IoT Control Center and various carrier-side LMF implementations, ensuring a consistent data format regardless of the local hardware vendor.

Q D4: How are connectivity rural "Dead Zones" managed?

A: Spatial Truth ensures a continuous "**source of truth**" by blending three distinct layers into a single coordinate stream:

- **Urban:** 5G + Wi-Fi RTT (Passive ranging).
- **Highway:** 5G + V2X (Safety bands).
- **Rural/Remote:** NTN (Satellite bridge via Skylo).